

ADVANCED EXCEL REFERENCE • 2024 EDITION

Master Excel's Most Powerful Tools

A structured reference covering data integrity, visual formatting, and lookup functions — from basics to advanced patterns.

5

Topics Covered

30+

Formula Examples

50+

Use Cases

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01 — DATA VALIDATION

Control What Enters Your Cells

Restrict data type, range, or format — prevent errors before they happen and keep spreadsheets clean and consistent.

VALIDATION TYPES

Type	Description	Example Setting
Whole Number	Allow only integers	Between 1 and 100
Decimal	Allow decimal values	Greater than 0.5
List	Dropdown from predefined list	=A1:A10 or Male,Female
Date	Restrict to date range	Between 01/01/2020 and today

Time	Valid time values only	Between 09:00 and 17:00
Text Length	Limit character count	<= 50 characters
Custom Formula	Any TRUE/FALSE formula	=ISNUMBER(A1)

CUSTOM FORMULA EXAMPLES

=ISNUMBER(A1) Cell must contain a number only	=LEN(A1)<=10 Text length must not exceed 10 characters
=COUNTIF(\$B\$2:\$B\$100,B2)=1 Prevent duplicate entries in column B	=AND(A1>=1, A1<=100) Number must be between 1 and 100
=ISBLANK(A1)=FALSE Cell must not be empty	=MOD(A1,5)=0 Only multiples of 5 are allowed
=NOT(ISNUMBER(SEARCH(" ",A1))) No spaces allowed in text entry	=ISODD(A1) Only odd numbers permitted

» **Named Ranges for Dropdowns**

Use a Named Range for your dropdown source (e.g. DeptList). Enter =DeptList in the Source box. When the named range updates, all dropdowns refresh automatically — no formula edits needed.

WHEN TO USE

Data-entry forms requiring strict formats	Data is imported automatically via formulas
Dropdown menus for consistent categorisation	Values are computed by other cells
Preventing invalid dates in financial models	Sheet is read-only / display only
Enforcing unique entries in ID or SKU columns	

02 — CONDITIONAL FORMATTING

Let Your Data Speak Visually

Apply colors, icons, and data bars automatically — transform raw numbers into instant visual insights without changing underlying data.

RULE CATEGORIES

Category	Rules Available
Highlight Cells	Greater Than, Less Than, Between, Equal To, Text Contains, Duplicates
Top / Bottom	Top 10 Items, Bottom 10%, Above Average, Below Average
Data Bars	Gradient or solid fill bars proportional to cell value
Color Scales	2- or 3-color gradient spread across a range of values
Icon Sets	Arrows, traffic lights, stars and flags based on thresholds
Custom Formula	Any formula returning TRUE/FALSE — the most powerful option

CUSTOM FORMULA EXAMPLES

=C2>1000 Highlight entire row where column C exceeds 1000	=MOD(ROW(),2)=0 Alternate row shading — zebra stripes
=TODAY()-B2>30 Flag items older than 30 days	=AND(C2>0,D2="Pending") Highlight rows: sales > 0 AND status is Pending
=WEEKDAY(A2,2)>5 Highlight weekend dates (Saturday/Sunday)	=RANK(B2,B\$2:B\$100)<=3 Highlight top 3 values in column B
=C2/SUM(C\$2:C\$100)>0.1 Cells contributing more than 10% of total	=LEN(TRIM(A2))<>LEN(A2) Detect leading or trailing spaces in text

» Relative vs Absolute References in CF Rules

When formatting entire rows, anchor the column with \$ but leave the row relative: =C2>1000. This lets each row evaluate independently while always checking column C. Avoid CF on very large datasets (100k+ rows) — it can significantly slow Excel.

WHEN TO USE

Dashboard KPIs — red/amber/green traffic lights	Dataset exceeds 100,000+ rows
Sales data — highlight best and worst performers	Formatting must persist after CSV export
Project tracking — overdue tasks or deadlines	Logic is too complex for a formula rule
Data cleaning — blank cells, duplicates, outliers	

03 — LOOKUP FUNCTIONS

Find and Retrieve Data Anywhere

Search for a value in one dataset and pull related information from another — the backbone of any data-driven spreadsheet.

FUNCTION COMPARISON

Function	Direction	Complexity	Best Use Case	Key Limitation
VLOOKUP	Vertical ↓	Simple	Find by row key	Left-to-right only
HLOOKUP	Horizontal →	Simple	Find by col header	Top-to-bottom only
LOOKUP	Both	Simple	Sorted data search	Requires sorted data
INDEX	Both	Moderate	Return by position	Needs row+col number
MATCH	Both	Moderate	Find position only	Returns position, not value
INDEX+MATCH	Both	Moderate	Most flexible lookup	Two functions combined
XLOOKUP	Both	Simple	Modern replacement	Excel 365/2019+ only

MATCH FUNCTION

`=MATCH(lookup_value, lookup_array, [match_type])`

Returns the position of a value in a range

`=MATCH("Alice", A2:A100, 0)`

Exact match — returns row position of Alice

`=MATCH(500, B2:B50, 1)`

Largest value <= 500 (ascending sort required)

`=MATCH(500, B2:B50, -1)`

Smallest value >= 500 (descending sort required)

INDEX FUNCTION

`=INDEX(array, row_num, [col_num])`

Returns value at a specific row/column intersection

`=INDEX(A1:D100, 5, 3)`

Value from row 5, column 3 of A1:D100

`=INDEX(B2:B100, 10)`

10th value from a single-column range

INDEX + MATCH — THE POWER COMBINATION

`=INDEX(C2:C100, MATCH("Alice", A2:A100, 0))`

Find Alice in col A, return her salary from col C

```
=INDEX(A2:D100, MATCH("Q3", A2:A100, 0), MATCH("Revenue", A1:D1, 0))
```

2D lookup — find row by Q3, column by Revenue header

```
=INDEX($B$2:$B$100, MATCH(MAX($C$2:$C$100), $C$2:$C$100, 0))
```

Name of the person with the highest score

» Why INDEX+MATCH beats VLOOKUP

INDEX+MATCH can look left, is unaffected by column insertions, and searches only one column rather than scanning the full table array — significantly faster on large datasets. Consider it your default for any complex lookup.

04 — VLOOKUP

Vertical Lookup

Search down a column and return a value from the same row. The most widely used lookup function in Excel.

SYNTAX

```
=VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])
```

Core syntax

PARAMETERS

Parameter	Required?	Description
lookup_value	Required	The value to find (text, number, or cell reference)
table_array	Required	Range containing data — first column must hold lookup values
col_index_num	Required	Column number in table_array to return (1 = first column)
range_lookup	Optional	FALSE = exact match (recommended) TRUE = approximate match

EXAMPLES — BASIC TO ADVANCED

```
=VLOOKUP(A2, $D$2:$G$100, 2, FALSE)
```

Exact match: look up A2 in D2:G100, return column 2

```
=VLOOKUP("EMP001", EmployeeTable, 3, FALSE)
```

Look up employee ID in named range, return 3rd column (dept)

```
=IFERROR(VLOOKUP(A2,$D$2:$G$100,2,FALSE), "Not Found")
```

Show Not Found instead of #N/A error

```
=IFERROR(VLOOKUP(A2,Sheet2!$A:$D,3,FALSE), "-")
```

Cross-sheet VLOOKUP — pulls data from Sheet2

```
=VLOOKUP(A2&B2, $D$2:$F$100, 3, FALSE)
```

Multi-criteria: concatenate two keys (requires helper column)

```
=VLOOKUP(D2, $A$2:$B$6, 2, TRUE)
```

Approximate match — grade brackets, tax ranges (data sorted asc)

COMMON ERRORS & FIXES

Error	Cause	Fix
#N/A	Value not found in first column	Use IFERROR(); check for spaces with TRIM()
#REF!	col_index_num exceeds table width	Count columns; col_index must be <= table width
#VALUE!	col_index_num is less than 1	col_index must be a positive integer
Wrong result	range_lookup=TRUE on unsorted data	Always use FALSE for exact match

» Always use absolute references (\$)

Lock your table_array with \$D\$2:\$G\$100. When copying formulas down, relative references shift — causing wrong results or #REF! errors. Press F4 after selecting the range to toggle absolute referencing.

05 — HLOOKUP

Horizontal Lookup

Search across the first row and return a value from a specified row below. Ideal when your headers run horizontally across columns.

SYNTAX

```
=HLOOKUP(lookup_value, table_array, row_index_num, [range_lookup])
```

Core syntax

PARAMETERS

Parameter	Required?	Description
lookup_value	Required	Value to search for in the FIRST ROW of table_array
table_array	Required	Range — first row must contain the lookup header values
row_index_num	Required	Which row number to return (1 = header row itself)
range_lookup	Optional	FALSE = exact match TRUE = approximate (sorted first row)

EXAMPLE DATASET — MONTHLY SALES BY REGION

Region	Jan	Feb	Mar	Q1 Total
North	45,200	51,300	48,700	145,200
South	38,900	42,100	55,600	136,600
East	62,400	58,200	70,100	190,700
West	29,800	33,500	37,200	100,500

EXAMPLES

```
=HLOOKUP("Feb", $A$1:$E$5, 3, FALSE)
```

South's Feb sales = 42,100 (row 3 = South)

```
=HLOOKUP("Q1 Total", $A$1:$E$5, 4, FALSE)
```

East's Q1 Total = 190,700 (row 4 = East)

```
=HLOOKUP("Bonus", $A$1:$L$8, MATCH(A2, $A$1:$A$8, 0), FALSE)
```

Combine MATCH for a dynamic row_index_num

```
=IFERROR(HLOOKUP(A1, $B$1:$Z$10, 2, FALSE), 0)
```

Return 0 instead of #N/A if column header not found

```
=HLOOKUP("Target", Summary!$A$1:$M$6, 5, FALSE)
```

Cross-sheet HLOOKUP referencing the Summary tab

» When HLOOKUP makes sense

HLOOKUP fits financial models where time periods (Jan, Feb, Mar...) are column headers and metrics are rows. If you use HLOOKUP frequently, consider whether transposing your data to a vertical layout would simplify future analysis.

06 — COMPARISON & DECISION GUIDE

Choose the Right Tool Every Time

A quick-reference comparison across all four lookup methods, plus a decision flowchart so you always pick the correct function.

FUNCTION FEATURE COMPARISON

Feature	VLOOKUP	HLOOKUP	INDEX+MATCH	XLOOKUP
Search Direction	Vertical ↓	Horizontal →	Both ↔	Both ↔
Lookup Column Pos.	Must be 1st	Must be 1st row	Any position	Any position
Return Direction	Right only →	Down only ↓	Any direction	Any direction
Multiple Results	First only	First only	Array capable	Built-in
If Not Found	IFERROR needed	IFERROR needed	IFERROR needed	Built-in param
Performance	Moderate	Moderate	Fast	Fast
Excel Version	All versions	All versions	All versions	365/2019+ only
Ease of Use	Easy	Easy	Moderate	Easy-Moderate

DECISION FLOWCHART

Data is vertical AND lookup column is first?	→	VLOOKUP
Data is horizontal AND lookup row is first?	→	HLOOKUP
Lookup column is NOT first / need to look left?	→	INDEX+MATCH
Need built-in not-found handling (Excel 365)?	→	XLOOKUP
Complex 2-way lookup (row AND column)?	→	INDEX+MATCH 2D
Simple range lookup on sorted data?	→	VLOOKUP TRUE

XLOOKUP — THE MODERN STANDARD (EXCEL 365+)

```
=XLOOKUP(lookup_value, lookup_array, return_array, [not_found], [match_mode], [search_mode])
```

Full syntax — most parameters are optional

```
=XLOOKUP("Alice", A2:A100, C2:C100)
```

Find Alice in col A, return salary from col C

```
=XLOOKUP("Alice", A2:A100, B2:D100)
```

Return multiple columns at once — not possible with VLOOKUP

```
=XLOOKUP(A2, A2:A100, B2:B100, "Not Found")
```

Built-in not-found — no IFERROR wrapper needed

=XLOOKUP(A2, C2:C100, B2:B100)

Left lookup — return_array can be LEFT of lookup_array

BEST PRACTICES SUMMARY

OK	Always use exact match (FALSE/0)	Approximate match silently returns wrong results if data is not perfectly sorted ascending.
OK	Lock table references with \$	Use \$A\$2:\$D\$100 — formulas shift when copied, causing incorrect lookups.
OK	Wrap lookups in IFERROR	=IFERROR(VLOOKUP(...), "Not Found") prevents ugly #N/A errors in reports.
OK	Use Named Ranges	Name your lookup tables (e.g. PriceTable) for readable, maintainable formulas.
OK	Prefer INDEX+MATCH for large data	It searches only one column/row vs VLOOKUP which scans the entire table array.
OK	Migrate to XLOOKUP when possible	Simpler syntax, built-in error handling, supports left lookups and multiple results.

» Future-proofing: start migrating to XLOOKUP

Microsoft officially recommends XLOOKUP as the replacement for both VLOOKUP and HLOOKUP in Excel 365. If your organisation uses Excel 365 or Excel 2021+, start adopting it now — simpler syntax, built-in error handling, and no wrapper functions required.